

Example 1a: Find x

Example 1a

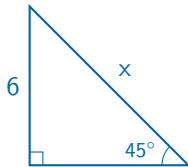
Example 1b

Step-by-Step Solution

① Identify Triangle Type:

$$90^\circ - 45^\circ = 45^\circ.$$

This is a 45° - 45° - 90° triangle.



Example 1a: Find x

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Example 1b

Step-by-Step Solution

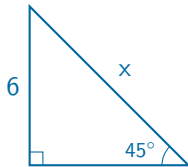
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This is a 45° - 45° - 90° triangle.

② Relationship:

$$\text{Hypotenuse} = \text{Leg} \cdot \sqrt{2}$$



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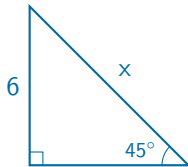
② Relationship:

$$\text{Hypotenuse} = \text{Leg} \cdot \sqrt{2}$$

③ Substitute and Solve:

$$\text{Leg} = 6$$

$$x = 6\sqrt{2}$$



Example 1b: Find x

Example 1a

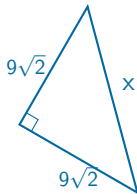
Example 1b

Step-by-Step Solution

1 Identify Triangle Type:

Legs are congruent (Leg = $9\sqrt{2}$).

It is a 45° - 45° - 90° triangle.



Example 1b: Find x

Example 1a

Example 1b

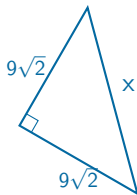
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Example 1b: Find x

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Example 1b

Step-by-Step Solution

① Identify Triangle Type:

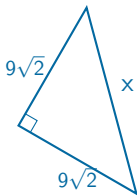
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Hypotenuse = Leg $\cdot \sqrt{2}$

③ Substitute Value:

$$x = (9\sqrt{2}) \cdot \sqrt{2}$$



Example 1b: Find x

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Example 1b

Step-by-Step Solution

1 Identify Triangle Type:

Legs are congruent (Leg = $9\sqrt{2}$).
It is a 45° - 45° - 90° triangle.

2 Relationship:

Hypotenuse = Leg $\cdot \sqrt{2}$

3 Substitute Value:

$$x = (9\sqrt{2}) \cdot \sqrt{2}$$

4 Simplify Answer:

$$x = 9 \cdot (\sqrt{2} \cdot \sqrt{2})$$

$$x = 9 \cdot 2 = 18$$

